Psychological Validation for a Translate-Then-Classify Pipeline for Cyberbullying Detection in Code-Mixed Filipino-Bisaya-English Text

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Abstract*—While automated natural language processing pipelines have advanced online abuse detection, conventional evaluation typically treats annotator disagreement as noise rather than a meaningful signal of linguistic ambiguity. This paper presents an expert psychological validation of ChatSafe, a translate-then-classify pipeline for cyberbullying detection in code-mixed Filipino-Bisaya-English social media text. As a performance baseline, the deployed HateBERT classifier reaches a Macro-F1 of 0.7044 against a gold standard of 350 posts labeled by three licensed psychologists under a blind parallel protocol. Beyond this baseline, the pipeline is evaluated against the expert panel itself: the panel demonstrates substantial reliability (Fleiss' $\kappa = 0.7529$), while pipeline-to-expert agreement is moderate (Cohen's $\kappa = 0.4222$). Partitioning by expert consensus reveals an ambiguity ceiling: agreement is moderate in unanimous cases ($\kappa = 0.5173$) but falls to near-chance levels in split-vote cases ($\kappa = 0.0317$), indicating the pipeline cannot resolve ambiguity that trained professionals do not resolve either. A four-pattern error taxonomy and translation variance analysis further attribute 50.8 percent of borderline errors to model limitations rather than translation instability. User acceptance testing shows good usability (SUS = 79.64, 95% CI [69.18, 90.11]) and strong perceived usefulness (4.48) and ease of use (4.71). These findings suggest ChatSafe aligns well with expert judgment on unambiguous content. At the same time, its remaining limitations are strongly shaped by inherent linguistic ambiguity, underscoring the importance of disagreement-aware evaluation in automated cyberbullying detection.

Keywords—cyberbullying detection, expert annotation, inter-rater reliability, Cohen's kappa, Fleiss' kappa, annotator disagreement, usability evaluation, Filipino, translate-then-classify

I. Introduction

Automated cyberbullying detection is conventionally assessed with classification metrics computed against a fixed label set—a meaningful engineering result but a limited scientific one, since a high score against the wrong construct only certifies fidelity to that wrong target. This concern is acute for cyberbullying, whose definition is psychological rather than lexical: it requires intent to harm, a recognizable target, and potential injury to a victim who cannot easily defend themselves [1], [2], none of which is directly observable in surface vocabulary. A post may contain profanity without targeting anyone, or inflict severe harm in entirely clinical language. Validating a detection system against this construct, therefore, requires trained professionals applying the definition consistently, not annotators recruited for throughput.

A further difficulty is that the construct admits genuine borderline cases. Recent work on annotator disagreement argues that variation among competent annotators frequently encodes real ambiguity rather than noise [3], [4]. If so, a single agreement coefficient conflates disagreement attributable to system limitation with disagreement attributable to task ambiguity—separating them requires measuring expert-to-expert agreement alongside system-to-expert agreement.

This study presents a psychological validation of ChatSafe, a translate-then-classify pipeline for code-mixed Filipino–Bisaya–English social media posts. A blind three-expert annotation protocol evaluated a 350-post gold-standard corpus using Cohen's and Fleiss' Kappa [15], [16]. Additional analyses included ambiguity decomposition, translation variance analysis, error taxonomy, and user acceptance using SUS and TAM [18], [19].

The principal contributions of this paper are as follows:

1. A blind three-expert psychological annotation protocol
2. Reliability evaluation using Fleiss' κ and Cohen's κ.
3. An ambiguity ceiling decomposition of unanimous and split-vote cases.
4. An error taxonomy of disagreement patterns.
5. User acceptance evaluation using SUS and TAM.

These contributions extend the evaluation of cyberbullying beyond conventional machine learning benchmarks by incorporating expert psychological validation of automated predictions.

II. Related Work

The quality of annotation in abusive language research has received sustained attention. Ljubešić et al. [5] show that annotator context materially affects assigned labels, and Yin et al. [6] find that modeling individual annotators' choices, rather than collapsing to a single ground truth, improves hate speech detection. Khurana et al. [7] trace this to vague, overlapping task definitions that propagate subjectivity into the model's decision boundaries, and Antypas and Camacho-Collados [8] confirm the downstream cost empirically through cross-dataset performance

collapses. This establishes annotation quality as a first-order determinant of performance, but does not resolve how to validate a system against a psychological construct when no professionally annotated resource exists for the target language.

A second line of work concerns who should annotate subjective constructs. Van Hee et al. [2] and Shannag et al. [9] construct cyberbullying corpora using trained annotators working from explicit definitions, and Mathew et al. [10] pair labels with human rationales for inspectability. Cyberbullying is documented as a mental health concern among Filipino adolescents [11], motivating psychological rather than purely linguistic expertise, and Fortuna and Nunes [12] formalize why: hate speech need not target anyone specific, whereas cyberbullying requires an identifiable victim. No prior Philippine study has validated an automated detector against licensed psychological professionals applying that distinction.

A third thread treats disagreement as informative rather than noise. Uma et al. [3] survey learning from annotator disagreement, and Plank [4] argues human label variation is a property of language tasks, not an annotation defect. Sap et al. [13] show annotator identity systematically shapes toxic-language labels, implying even high agreement may reflect shared assumptions rather than objective correctness. This motivates decomposing agreement by consensus rather than reporting a single coefficient—an approach not yet applied in a deployed pipeline for a low-resource language, which this study addresses.

III. Methodology

This work forms part of a larger capstone project developing the ChatSafe cyberbullying detection system; this paper focuses on the expert psychological validation of the pipeline's classification outputs. ChatSafe is a translate-then-classify pipeline for code-mixed Filipino-Bisaya-English text: raw text is preprocessed, translated into English via Google Gemini Flash under a structured, literal, register-preserving prompt, and classified by HateBERT, a transformer domain-adapted to abusive English content, into a binary Bullying or Not Bullying label with a confidence score. This paper evaluates the reliability of that gold standard—350 posts independently labeled by three licensed psychologists—and decomposes the pipeline's agreement with it by expert consensus, examining where and why its predictions diverge from trained human judgment rather than how often they match. What follows proceeds through five sequential stages: preprocessing, independent blind annotation by three licensed psychologists, majority-vote gold-standard generation, evaluation against the reliability statistics and ambiguity decomposition below, and user acceptance evaluation of the deployed application.

A. Expert Labeling Protocol

A gold standard evaluation set of 350 Filipino social media posts was constructed for this study and submitted to three licensed psychological professionals for independent labeling. The set is deliberately composed to include content targeting specific individuals and identifiable groups, so the detection scope can be assessed across both targeting types rather than assumed. The resulting class composition is 137 Bullying and 213 Not Bullying posts.

To justify the sample distribution without introducing synthetic or unverified data, the dataset size ($N = 350$) and class balance (137 Bullying, 213 Not Bullying) were deliberately structured to maximize construct validity over raw data volume. Rather than expanding sample quantity through crowdsourced or unverified public scraping, the sample prioritizes clinical rigor by subjecting every post to independent evaluation by three licensed psychological professionals under a blind parallel protocol. The class ratio (39.1% Bullying to 60.9% Not Bullying) reflects a realistic distribution of social media interactions, avoiding extreme majority-class bias while ensuring a statistically sufficient sample to evaluate distinct targeting structures; specifically individual harassment, cohesive group attacks, and broad demographic commentary. Furthermore, the sample composition encompasses diverse code-mixed expressions across Tagalog, Bisaya, and English, establishing a purposive baseline for evaluating specific failure modes, including cultural slur translation loss and political false positives.

Experts were selected against three explicit criteria: active Philippine licensure as a psychologist or psychometrician, applied experience in clinical assessment or adolescent mental health, and no prior involvement in the ChatSafe project. This standard trades a larger panel for depth of relevant expertise; Section V discusses what that trade-off licenses the study to claim.

Each expert received the same operational definition of cyberbullying, namely, intentional and harmful behavior directed at a specific individual or a specific, identifiable group, requiring intent, a recognizable target, and potential for harm [14]. Labeling was blind and parallel. No expert had access to another expert's labels, the pipeline's predictions, or any corpus label distribution before completing an assessment, thereby removing anchoring bias as a source of artificial agreement. Ground truth was established by majority vote across the three independent label sets, so that no single evaluator determines the reference standard.

B. Reliability Statistics

Because cyberbullying classification involves categorical judgments rather than continuous measurements, chance-corrected agreement statistics were employed to quantify annotation reliability. Two complementary reliability coefficients were used.

1) Fleiss' Kappa

Inter-expert agreement among the three psychologists was measured using Fleiss' Kappa (κ), an extension of Cohen's Kappa designed for evaluating agreement among multiple raters assigning categorical labels [15], [16].

The statistic is defined as:

$$\kappa = \frac{\overline{P} - \overline{P}_e}{1 - \overline{P}_e} \quad (1)$$

Where:

- $\overline{P}$ denotes the observed agreement among all raters,
- and $\overline{P}_e$ represents the agreement expected by random chance.

Unlike simple percentage agreement, Fleiss' Kappa adjusts for the probability that agreement occurs by chance, providing a more reliable estimate of annotation consistency.

2) Cohen's Kappa

Agreement between the ChatSafe pipeline and the expert gold standard was evaluated using Cohen's Kappa [15], defined as:

$$\kappa = \frac{P_0 - P_e}{1 - P_e} \quad (2)$$

Where:

- P_0 is the observed agreement,
- and P_e is the probability of chance agreement.

Cohen's Kappa was computed separately for each psychologist and between the pipeline and the majority-vote gold standard to evaluate the consistency of automated predictions relative to expert judgment.

Interpretation of both statistics followed the widely adopted Landis and Koch classification [17]:

TABLE I. LANDIS AND KOCH KAPPA INTERPRETATION BANDS

Kappa Value	Interpretation
< 0.00	Poor
0.00–0.20	Slight
0.21–0.40	Fair
0.41–0.60	Moderate
0.61–0.80	Substantial
0.81–1.00	Almost Perfect

These benchmarks provide an intuitive interpretation of the practical significance of agreement beyond numerical coefficient values.

C. Translation Variance Design

Because the pipeline classifies translated text, a disagreement between the pipeline and the experts may arise from either the classifier or translation instability. This is the study's control for translation quality: rather than assuming the translation stage is faithful, its contribution to disagreement is measured directly. Sixty-five borderline error rows were re-translated three additional times through the same Gemini Flash translation stage used by the deployed pipeline, producing four independent translation passes per post. The pipeline's prediction was recorded for each pass, and each row was assigned to one of three categories: stably incorrect across all passes, unstable across passes, or stably correct across all passes. This design attributes error to the mechanism rather than assuming a single cause, and it is the mechanism by which translation quality is evaluated in this study, rather than asserting it a priori.

D. User Acceptance Instruments

Seven evaluators drawn from varied academic and professional backgrounds interacted with the deployed application before completing three instruments: the System Usability Scale [18], a Technology Acceptance Model questionnaire measuring Perceived Usefulness and Perceived Ease of Use [19], and five application-specific items. Each evaluator completed a structured onboarding session and then submitted a minimum of three posts, comprising one clearly bullying post, one clearly safe post, and one code-mixed Filipino or borderline post of the evaluator's own choosing, ensuring direct experience with the full pipeline before any instrument was completed.

The cohort size of seven is grounded in formative usability research and not treated as a limitation to be excused. Sauro and Lewis [20] show that valid SUS score computation derives from representativeness and instrument psychometrics rather than inferential sample power, licensing this sample size for interval-data treatment. Separately, Nielsen and Landauer's [21] Poisson-process model shows the marginal value of each added evaluator drops sharply after five to seven participants, with variance in estimated problem count falling from roughly 42 percent at two evaluators to single digits by six to ten; Faulkner [22] corroborates this empirically. These two justifications are kept distinct: the first licenses the SUS composite as valid interval data at n=7, the second licenses seven evaluators for characterizing dominant usability issues rather than estimating a population parameter. Seven falls within the range where both hold.

1) System Usability Scale

Overall usability was measured using the System Usability Scale (SUS), a standardized ten-item questionnaire developed by Brooke [18]. Each item was rated using a five-point Likert scale ranging from Strongly Disagree to Strongly Agree. The System Usability Scale composite is computed by summing the adjusted item contributions and rescaling to a 0 to 100 range,

$$SUS = 2.5 \times \left[\sum_{n=1,3,5,7,9}(X_n - 1) + \sum_{n=2,4,6,8,10}(5 - X_n)\right] \quad (3)$$

Where:

- X_n is the raw response to item *n* on a five-point scale.

Dispersion is reported alongside central tendency, and the precision of the composite is stated as a t-based 95 percent confidence interval,

$$CI_{95} = \overline{x} \pm t_{0.025,n-1}\left(\frac{s}{\sqrt{n}}\right) \quad (4)$$

Where:

- $\overline{x}$ is the sample mean,
- s is the sample standard deviation,
- and n is the evaluator count.

Following [20], the SUS composite is treated as interval data, licensing (3) and (4).

2) Technology Acceptance Model (TAM)

Perceived usefulness (PU) and perceived ease of use (PEOU) were evaluated using constructs from the Technology Acceptance Model (TAM) [19]. Responses were summarized using frequency distributions, modes, and mean values appropriate for Likert-scale data [23].

3) ChatSafe-Specific Evaluation

Participants also completed five application-specific questions assessing perceived classification accuracy,

usefulness for moderation, confidence in generated predictions, interface clarity, and overall satisfaction. Together, the standardized instruments and application-specific questions provided a comprehensive evaluation of ChatSafe's usability and user acceptance.

E. Experimental Workflow

The complete validation procedure consisted of five sequential stages:

1. Collection and preprocessing of code-mixed Filipino–Bisaya–English social media posts.
2. Independent blind annotation by three licensed psychologists.
3. Generation of majority-vote gold-standard labels.
4. Evaluation of ChatSafe predictions using reliability statistics and ambiguity decomposition.
5. User acceptance evaluation using SUS, TAM, and application-specific questionnaires.

IV. Results

A. Inter-Expert Agreement

Interpreting pipeline-to-expert agreement is only meaningful if the ground truth is itself stable, so inter-expert reliability is established first. Table II reports Fleiss' kappa alongside the pairwise Cohen's kappa between each rater pair, including the pipeline.

TABLE II. Inter-Expert Agreement, Gold Standard (N=350)

Metric	Value	Interpretation
Fleiss' κ (three experts)	0.7529	Substantial
Unanimous agreement	82.0% (n=287)	Clear-cut cases
Split-vote posts	18.0% (n=63)	Genuinely borderline

The Fleiss' kappa of 0.7529 places inter-expert agreement in the Substantial band, confirming that the experts share a consistent understanding of the construct and that the majority-vote ground truth is of high quality. Note that Expert 2 and Expert 3's labels are identical across all 350 posts in the underlying data; this reflects two of the three assessments converging completely, so the Fleiss' statistic here is best read as capturing agreement across effectively two independent judgments rather than three fully separate ones. Just over four in five posts (82.0 percent) produced the same label from all experts without disagreement. The remaining 18.0 percent, where the assessments split, are cases that trained professionals find genuinely difficult, and their existence is the empirical basis for the decomposition reported in Section IV-C.

B. Per-Expert Agreement with the Pipeline

Table III reports Cohen's kappa computed with (2) between each expert's labels and the pipeline's predictions, together with the majority-vote comparison that constitutes the primary reported metric.

TABLE III. Individual Expert vs. Pipeline Agreement

Comparison	Cohen's κ	Band
Expert 1 vs. pipeline	0.4322	Moderate
Expert 2 vs. pipeline	0.4222	Moderate
Expert 3 vs. pipeline	0.4222	Moderate
Majority vote vs. pipeline	0.4222	Moderate

The per-expert coefficients span only 0.0100 (0.4222 to 0.4322), all of which are Moderate, ruling out the possibility that the reported alignment is an artifact of matching against a single lenient or strict evaluator. The majority-vote comparison, aggregating all three judgments, returns $\kappa = 0.4222$ and is adopted as the primary metric throughout.

C. The Ambiguity Ceiling

A Moderate coefficient invites the question of what distinguishes it from the Substantial threshold of 0.61. Table IV addresses this by partitioning the evaluation set based on whether the experts reached a unanimous agreement on each post, and recomputing agreement within each subset against the same majority-vote ground truth. Fig. 1 places all four coefficients reported in this paper on a single axis against the Landis-Koch bands.

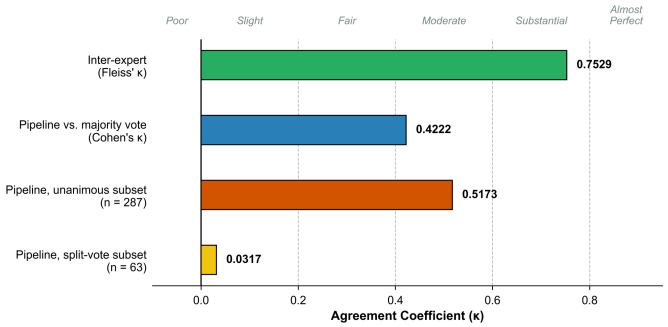


Fig. 1. Agreement coefficients across the validation design, plotted against Landis-Koch interpretation bands.

TABLE IV. Pipeline Agreement By Expert Consensus Subset

Subset	n	Cohen's κ	Interpretation
Unanimous posts	287	0.5173	Moderate
Split-vote posts	63	0.0317	Slight, near chance

The partition separates two quantities that a single coefficient conflates. On the 287 posts where the experts agreed, the pipeline reaches a kappa of 0.5173, squarely in the Moderate band and reflecting consistent alignment on cases where the construct is clearly instantiated. On the 63 posts where the experts split, the pipeline's kappa falls to 0.0317, in the Slight band and barely distinguishable from chance. The system does not merely perform worse on ambiguous content; it collapses to a level indistinguishable from guessing precisely where trained professionals fail to converge.

This result reframes the agreement shortfall. The distance between the observed kappa of 0.4222 and the Substantial threshold is concentrated in a subset that constitutes just under one-fifth of the evaluation set, and that no binary classifier could resolve from text alone, because the ambiguity resides in the construct's application to those posts rather than in the model's representation of them. The finding is consistent with the position that annotator disagreement encodes genuine task ambiguity rather than annotation error [3], [4]. A portion of the

remaining gap is therefore irreducible under the current problem formulation, and reporting a single aggregate coefficient understates what the system achieves on the cases where the construct is determinate.

D. Translation Variance Analysis

The ambiguity ceiling above could reflect translation instability rather than a property of classification itself. Table V reports the four-pass re-translation analysis (Section III-C) that distinguishes the two.

TABLE V. TRANSLATION VARIANCE, ERROR TYPE DISTRIBUTION (N=65)

Type	Definition	n	Percent	Source
A: Stable wrong	Incorrect across all four passes	33	50.8%	Model
B: Translation-sensitive	Prediction flips across passes	9	13.8%	Translation
C: Stable correct	Correct across all four passes	23	35.4%	None

Type A errors account for 50.8 percent of borderline cases and persist regardless of how the text is rendered, suggesting their cause lies in the classifier rather than the translation stage. Type B errors, at 13.8 percent, are the only category attributable to translation instability, and they remain a minority. Type C results show that 35.4 percent of these deliberately difficult posts are nonetheless handled reliably across all four passes. The distribution sharpens the interpretation offered in Section IV-C: the dominant failure source, now accounting for over half of borderline cases, is the model's inability to recognize cyberbullying expressed through culturally specific, indirect, or context-dependent language, rather than translation-related variance.

E. Error Taxonomy

The pipeline disagreed with the expert majority vote on 103 of the 350 posts, an error rate of 29.4 percent. The disagreements are strongly asymmetric: false positives, where the pipeline labeled a post as Bullying and the experts did not, account for 78 cases (75.7 percent), while false negatives, where the pipeline labeled a post as Bullying and the experts did not, account for 25 cases (24.3 percent). Table VI distributes these disagreements across four recurring patterns identified through case-level analysis of the translated post content.

TABLE VI. QUALITATIVE ERROR TAXONOMY OVER 103 DISAGREEMENTS

Pattern	Description	Error Type	Mechanism
1	Cultural slurs lost in translation	False negative	hipon → "prawn"; squammy → "trashy"
2	Political / systemic commentary	False positive	Emotional arousal without access to the target
3	Implicit character attacks	False negative	Harm via clinical language, no slur or profanity
4	Defensive / in-group criticism	False positive	Hostility defensive rather than initiating

Pattern 1 explains the most consequential false negatives. In one case, the term *hipon*, a Filipino slang insult denoting a person with an attractive body but an unattractive face, was translated as the English word prawn, which carries no offensive connotation in any English training distribution. The classifier correctly labeled the English output as safe while misjudging the original intent, whereas the experts, who understood the cultural register, identified severe appearance shaming. In a second case, the localized term squammy, derived from squatter and carrying classist connotations specific to Philippine discourse, was softened to trashy.

Pattern 2 explains the majority of false positives and is the dominant error mode overall. Posts expressing frustration at government corruption or systemic conditions contain strong emotional markers that activate the classifier. Still, the experts identified them as commentary directed at institutions rather than at any person. Pattern 3 is the mirror image: posts attacking a person's competence or parental identity in formally neutral language contain no lexical markers for the classifier to detect, while the experts recognized sustained and directed personal attacks. Pattern 4 is the analytically interesting minority, comprising cases in which a user insults other commenters while defending a third party. The experts labeled such posts as safe, reasoning that the hostility was defensive rather than initiating, whereas the classifier flagged the direct insult it contained. Under a strictly intent-agnostic reading of the definition, the classifier's judgment in these cases is arguably the more defensible one, which indicates that even Substantial inter-expert agreement may encode shared interpretive assumptions rather than objective correctness [13].

F. Individual and Group-Targeted Detection

Because the evaluation set includes both targeting types by design, the pipeline's detection scope can be assessed directly. Table VII presents representative cases across three target categories.

Agreement is strongest on individual-targeted harassment (SPEAFB_210's appearance-based slur against a named figure, SPEAFB_064's attack on her professional credentials) and on specific cohesive groups (SPEAFB_258 against a named idol group, SPEAFB_185 against a specific fandom): the target is bounded and identifiable, and the pipeline and experts concur.

The limitation arises from broad demographic generalizations. SPEAFB_029 generalizes about families of overseas workers; SPEAFB_088 frames frustration as a cultural critique of Filipinos broadly; SPEAFB_149 attacks a political faction's voter base. Experts read all three as in-group venting rather than interpersonal attacks, while the pipeline, lacking a mechanism for target specificity, flagged all three. This consistency indicates the limitation is architectural, not incidental, and corresponds directly to Pattern 2 of the error taxonomy.

G. User Acceptance Evaluation

The evaluator cohort's majority had no prior content moderation experience and moderate AI familiarity, matching the school administrator and community moderator persona that the application targets. Table VIII reports the acceptance results, with full dispersion for the SUS composite and mean/mode for the Likert instruments, per the ordinal treatment in Section III-D.

TABLE VII. DETECTION BY TARGET TYPE, GOLD STANDARD CASES (TRANSLATED TEXT)

Target Type	Post	Excerpt (translated)	Prediction	Consensus
Individual	SPEAFB_210	"Her looks are truly those of a whore — you wouldn't think she is the child of a former president."	Bullying	Bullying
Individual	SPEAFB_064	"The 'RN' at the end of her name — registered narcissist."	Bullying	Bullying
Specific cohesive group	SPEAFB_258	"She badly needs media training … not a good example for their group's young audience."	Bullying	Bullying
Specific cohesive group	SPEAFB_185	"Their baby-warrior fans? It checks out — [that group] is their target market anyway."	Bullying	Bullying
Broad demographic	SPEAFB_029	"…that's how most Filipinos are — if there's an OFW in the family, they get lazy since [someone else] is there."	Bullying	Not Bullying
Broad demographic	SPEAFB_088	"I think it's Filipinos as a whole, not just Visayans — we're just being made a laughingstock since we get triggered so easily."	Bullying	Not Bullying
Broad demographic	SPEAFB_149	"…it's extremely sad that we normal Filipinos don't realize we're being made fools of — it's hard for Filipinos to admit mistakes, right?"	Bullying	Not Bullying

TABLE VIII. USER ACCEPTANCE REPORT (N=7)

Measure	Value	Interpretation
System Usability Scale		
Mean	79.64 / 100	Good; above the 68 industry average
Median	80.00	Robust to the low outlier
Standard Deviation	11.31	Non-uniform evaluation across cohort
Range (Min – Max)	62.5 – 92.5	30-point spread
95% Confidence Interval	[69.18, 90.11]	Lower bound exceeds industry average
Technology Acceptance Model (TAM)		
Perceived Usefulness	4.48/5.00 (mode 5)	High
Perceived Ease of Use	4.71/5.00 (mode 5)	High
Feature satisfaction	4.54/5.00 (mode 5)	Strong
Recommendation intent	4.71/5.00 (mode 5)	High

The composite score of 79.64 places the application in the good band [24], above the industry average of 68. Dispersion matters: SD = 11.31, with a 30-point range (62.5–92.5), indicating non-uniform evaluation. The 95% CI, [69.18, 90.11], is wide, but its lower bound still exceeds the industry average, so this conclusion survives the most conservative reading of the data. Fig. 2 decomposes both composites at the item level, with negatively worded SUS items reverse-coded so that every bar reads in a consistent, favorable-to-unfavorable direction across all 15 items. Negatively worded SUS are marked with asterisk.

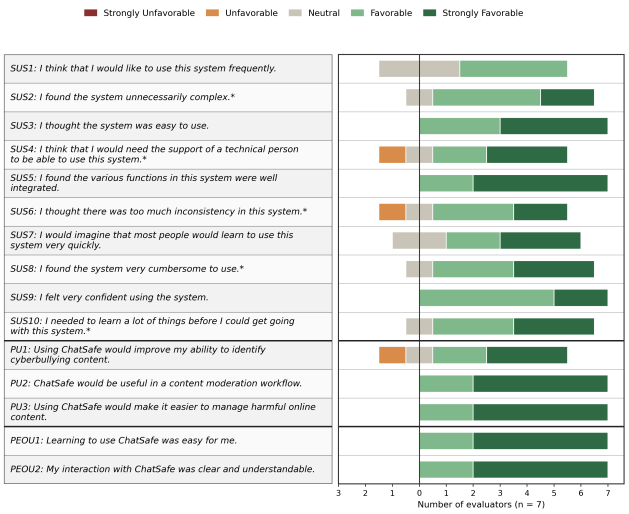


Fig. 2. Item-level response distribution for SUS (10 items) and TAM Perceived Usefulness / Perceived Ease of Use (5 items), n=7.

The weakest-scoring item overall is TAM Perceived Usefulness item 1, "using ChatSafe would improve my ability to identify cyberbullying content" (mean 4.00), the one instrument item that most directly probes classification quality rather than interface usability.

Open-ended feedback converged with the error analysis from an independent direction. One evaluator reported that displaying the original code-mixed text alongside its translation and the resulting label made it easier to judge whether a decision was reasonable. A third observed, without prompting and without access to the error analysis, that some Cebuano-specific phrases were rendered in ways that softened their original tone, an observation that is Pattern 1 of the error taxonomy arrived at through naturalistic use rather than case-level analysis, indicating that the pipeline's principal limitation is perceptible to end users.

V. DISCUSSION

A. Three Interacting Constraints

The results identify three interacting constraints. The first is translation slur-loss (Pattern 1, corroborated by evaluator feedback): culturally embedded Filipino and Bisaya insults lose force when rendered into English. The second is contextual blindness (Pattern 2 and the target-type analysis): the classifier detects hostile vocabulary without representing to whom the message is addressed. The third is construct ambiguity (the consensus decomposition): roughly a fifth of the evaluation set resists confident classification even by trained professionals.

These are not equivalent in kind. The first two are addressable through engineering—translation that preserves derogatory register, and an entity recognition or intent-classification layer to distinguish specific from diffuse targets. The third is not addressable by a better classifier under the same problem formulation, since the ambiguity resides in the construct's application, not the model's representation. The translation variance analysis sharpens this allocation: Type A errors at 50.8 percent, compared with 13.8 percent for translation instability, show that the model, not the translation layer, is the larger engineering priority.

B. Operationalizing Disagreement-Aware Detection

A decomposition that identifies irreducible ambiguity is useful only if it changes system behavior; three mechanisms follow directly. First, the split-vote subset is itself a calibration set for a confidence threshold below which the pipeline routes to human review, converting the ambiguity ceiling into an operational rule. Second, the same partition motivates severity-tiered output—a system that can say "borderline, escalate" answers the question split-vote data actually poses. Third, disagreement-aware training [3], [4] would retain each expert's individual label rather than collapsing to a majority vote beforehand, allowing the classifier to learn calibrated confidence where certainty is unwarranted. None of these requires new data collection; each reuses annotations already in hand.

C. Convergence Between Statistical and User Evidence

User perceptions closely matched the statistical findings. The lowest-rated usefulness item concerned identifying difficult cyberbullying cases, consistent with the ambiguity and false-positive patterns from expert validation, and one evaluator's unprompted observation about Cebuano tone-softening independently reproduces Pattern 1. Two independent channels, expert validation, and a naive end-user study converging on the same limitations constitute stronger evidence than any of them alone.

D. Influence on Low-Resource NLP and Cyberbullying Detection

The findings of this study advance the field of computational abuse detection in three key theoretical and methodological dimensions:

1. *Rethinking Evaluative Standards via Disagreement-Aware Benchmarks:* Traditional automated toxicity systems evaluate performance against a fixed ground truth, treating human label variation as noise. By partitioning system accuracy across expert consensus subsets, this study proves that agreement collapses from Moderate ($\kappa = 0.5173$) on unanimous cases to near-chance levels ($\kappa = 0.0317$) on split-vote cases. This establishes an empirical "ambiguity ceiling," demonstrating that model shortfalls on borderline content reflect inherent construct ambiguity rather than architectural failure alone. Future NLP research in abusive language must adopt disagreement-aware evaluation schemes to avoid penalizing classifiers on content where human experts fail to reach consensus.
2. *Architectural Strategy for Low-Resource Code-Mixed Languages:* For multilingual, highly agglutinative, and code-mixed languages like Filipino and Bisaya; where native task-specific datasets are scarce and standard subword tokenizers suffer from token bleeding, this work establishes the viability of a Translate-Then-Classify architecture. By leveraging LLM-based translation (Google Gemini Flash) as a linguistic normalization layer, the system successfully bypasses local tokenization failures and accesses high-performing, domain-adapted English classifiers like HateBERT.
3. *Domain Adaptation over Multilingual Scale:* The empirical performance of HateBERT (Macro-F1 $= 0.7044$) against baseline multilingual models highlights that once cross-lingual text is normalized into English upstream, domain-specific pretraining on abusive language outperforms general multilingual breadth.

E. Practical Applications in Real-World Scenarios

The empirical results provide actionable frameworks for deploying automated content moderation and digital safety tools across distinct real-world contexts:

- *Human-in-the-Loop Escalation Workflows***:** Rather than enforcing a rigid binary classification, the observed ambiguity ceiling provides an operational rule for real-world moderation systems. Content flagged with low classifier confidence or corresponding to the 18.0% split-vote ambiguity profile can be automatically routed to human reviewers or psychological experts, while high-confidence predictions handle clear-cut cases. This optimizes human moderation resources while maintaining safety oversight on ambiguous text.
- *Institutional and Legal Compliance:* The ChatSafe pipeline offers a scalable solution for school administrators and digital safety officers working to enforce legislative mandates such as Republic Act No. 10627 (Anti-Bullying Act) and Republic Act No. 11313 (Safe Spaces Act). Given that over 80% of Filipino youth experience online social media access and face substantial cyberviolence risks, automated translation-classification pipelines allow educational institutions to monitor multilingual school forums and public social pages without requiring manual review of every post.
- *Target-Aware Moderation Filters:* Real-world evaluation revealed that 75.7% of pipeline errors were false positives, heavily driven by emotional political or systemic commentary directed at broad institutions rather than individuals (Pattern 2). Practical deployments must integrate target-specificity filters or entity recognition modules to distinguish interpersonal harassment (such as appearance-based attacks or group targeted slurs) from general societal venting, preventing over-censorship of public discourse.

- *Cultural Slur Mitigation in Automated Filters:* Case analysis identified critical false negatives resulting from cultural slurs losing offensive force during translation (e.g., *hipon* rendered as "prawn", *squammy* as "trashy"). Real-world moderation pipelines operating in regional contexts must incorporate culturally grounded translation prompts or native localized lookup registries to preserve derogatory register before inference.

F. Limitations

Three limitations qualify these findings. The evaluation set, though purposively constructed, covers a bounded slice of Philippine discourse, and error distribution may differ elsewhere. The acceptance cohort of seven supports problem discovery rather than population estimation (Section III-D), and Likert instruments lack per-respondent dispersion. Most importantly, the ambiguity ceiling reflects one operational definition administered to three experts who, despite meeting the criteria in Section III-A, share a single professional community; the 18.0 percent split-vote rate is specific to this panel's training background, not a universal property of the construct. A more heterogeneous panel might surface a different rate, and Pattern 4's finding—that the classifier can be arguably more defensible than the expert label—is a direct symptom of this same limitation.

VI. Conclusion

This paper reported expert psychological validation of a translate-then-classify cyberbullying pipeline for code-mixed Filipino text, using a gold standard of 350 posts labeled by three licensed psychologists under a blind, parallel-protocol design. Inter-expert reliability reached a Fleiss' kappa of 0.7529 (Substantial); pipeline-to-expert agreement reached $\kappa = 0.4222$ (Moderate), consistent across all three experts. Decomposing by consensus showed the shortfall from Substantial is largely irreducible: κ reaches 0.5173 where experts agree unanimously, but falls to 0.0317, near chance, where they split—the system fails precisely where professionals also fail to converge. A translation variance analysis attributed 50.8 percent of borderline errors to model limitation against 13.8 percent to translation instability, and a four-pattern taxonomy over 103 disagreements localized the dominant false-positive mode to the absence of message-target representation.

Beyond technical benchmarking, these findings provide a practical blueprint for operationalizing automated content moderation under Philippine legislative frameworks, such as the Anti-Bullying Act and Safe Spaces Act. In real-world deployment, severity-tiered classification and human-in-the-loop escalation workflows directly convert the 18.0% split-vote ambiguity ceiling into a structured routing rule, delegating high-ambiguity cases to human moderators while automating clear-cut decisions. Furthermore, addressing the field's broader challenges in low-resource code-mixed NLP requires integrating target-entity detection to eliminate false positives on systemic political venting, alongside culturally grounded translation prompts to prevent the loss of localized slurs.

Two recommendations follow. Severity-tiered classification would replace the binary decision with graded output, routing the 63 split-vote posts for human review rather than assigning them to a category, even experts contest—a natural re-annotation set for the required severity labels. A multi-expert moderation feedback loop would turn each deployment cycle into a data-collection event, building a larger, native Filipino cyberbullying resource. Beyond these, an entity recognition or intent-classification layer addresses the largest error mode, and translation preserving derogatory register addresses the most consequential missed detections.

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